

Sunny Bhati

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Education

Indian Institute of Technology (IIT) Madras

B.TECH AND M.TECH IN ENGINEERING DESIGN (GPA 7.2/10)

Chennai, India

Aug. 2010 - May. 2015

Indian University (Bloomington)

M.S. IN COMPUTER SCIENCE WITH A CONCENTRATION IN ARTIFICIAL INTELLIGENCE (GPA 3.7/4.0)

Bloomington, Indiana, USA

Aug. 2019 - Apr. 2021

Research Project Experience

University of Maryland, Baltimore County

RESEARCH ASSISTANT

Baltimore, Maryland, USA

Feb. 2026 - Present

Pre-Doctoral Researcher

RESEARCH ASSISTANT, INDIAN INSTITUTE OF SCIENCE, BANGALORE

Indian Institute of Science,

Bangalore, India

Jan. 2024 - Dec. 2025

- Advised by **Prof. R. Venkatesh Babu**.
- **Project 1: 3D-Consistent Scene Generation from Monocular Videos using Multimodal Conditioning.** (Industrial Collaboration)
- Developed and evaluated methods for 3D-consistent scene generation from monocular videos with multimodal conditioning in collaboration with AISIN.
- Developed a multi-view consistent diffusion framework for 3D scene generation from monocular street-view videos with multimodal conditioning.
- Demonstrated improved downstream 3D object detection performance through synthetic data augmentation.
- **Project 2: Dataset Distillation and Coreset Methods.**
- Studied dataset distillation across soft- and hard-label regimes, showing that soft-label evaluation can hide true subset quality.
- Benchmarked large- and small-scale dataset distillation methods on ImageNet-1K against random and coreset baselines in hard-label settings.
- Proposed compute-aware methods, CAD-Prune and CA2D, to improve data-efficient learning on ImageNet-1K.
- Submitted this work to **CVPR 2026**, where it was accepted as an **ORAL** presentation.

Academic Projects

Automatic Image Matching and Transformations

FEATURE MATCHING, GEOMETRIC VISION, AND CLUSTERING

Academic Project

- Developed a framework to organize personal photo collections into coherent groups based on visual similarity.
- Implemented SIFT feature extraction with RANSAC-based geometric verification to estimate transformations and align images in a common coordinate system before clustering.

Optical Music Recognition from Scanned Documents

IMAGE PROCESSING AND PATTERN RECOGNITION

Academic Project

- Implemented 2D convolution from scratch with separable kernels on grayscale sheet-music scans.
- Detected musical symbols using Hamming distance and template-matching scores on Sobel edge maps.
- Applied Hough transform to detect staff lines, draw symbol-level bounding boxes, and tag textual note representations.

Publications

Underline denotes co-first authors or co-corresponding authors

INTERNATIONAL CONFERENCE

1. Priyam Dey, Sunny Bhati, Aditya Sahdev, Konda Reddy Mopuri, and R. Venkatesh Babu, “Rethinking Dataset Distillation: Hard Truths about Soft Labels,” in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026. **Accepted (ORAL)**.

Work Experience

Amazon Inc., Texas

Texas, U.S.A

APPLIED SCIENTIST

Jan. 2022 - Dec. 2023

- Built and deployed machine learning models for Amazon Search to improve ranking and relevance, contributing to key user engagement and business metrics.
- Designed multi-objective training labels based on user behavior, price sensitivity, and delivery speed, leading to an 8% improvement in click-through rate and a 0.6% increase in revenue; recognized through internal publication.
- Created a unified internal framework for data ingestion, model training, inference, and automated evaluation, cutting experiment latency from one week to 10 hours.
- Built feature pipelines to incorporate new signals into ranking models, and conducted experiments that improved model performance across key engagement metrics.

Cisco Inc, San Jose, CA

California U.S.A

SOFTWARE ENGINEER, MACHINE LEARNING

June 2021 - Jan. 2023

- Designed a sequential anomaly detection framework for data center network logs to identify failure precursors and predict switch-level faults.
- Embedded unstructured log sequences using Word2Vec and modeled temporal dependencies with an transformer model architecture for anomaly classification in network telemetry.

Senior Data Scientist

Qatar

ERNST AND YOUNG LLP, QATAR

Apr. 2017 - May. 2019

- Built statistical PD/LGD stress-testing models for financial institutions, incorporating scenario-based risk analysis for credit portfolio assessment.
- Performed credit model calibration and validation, including PD curve fitting, discriminatory power analysis, and qualitative review of rating model performance.
- Architected an ECL estimation system with an interactive R/SQL dashboard for analytics, reporting, and decision support.
- Awarded twice for innovation for delivering a production-grade end-to-end stress-testing pipeline.